UNCOVERING FINANCIAL FRAUD THROUGH SPATIOTEMPORAL ANALYSIS OF TRANSACTION DATA

By

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AUTHOR'S DECLARATION FOR ELECTRONIC SUBMISSION OF A MAJOR RESEARCH PROJECT (MRP)

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ABSTRACT

This Major Research Project studies the use of spatiotemporal analysis in financial fraud detection, with an emphasis on improving detection accuracy utilizing geographic and temporal characteristics. To replicate realistic fraud distribution patterns, a synthetic financial transactions dataset was augmented with real-world population data and mapped to key cities in the United States. Exploratory Data Analysis (EDA) was used to detect geographical and temporal patterns in fraudulent behavior. Advanced machine learning models, such as Logistic Regression, Random Forest, and XGBoost, were constructed and tested utilizing precision, recall, F1-score, and ROC-AUC metrics. XGBoost outperformed all other methods, indicating that it is ideal for incorporating spatiotemporal aspects into fraud detections systems. The findings emphasize the value of integrating geographic, temporal, and transactional data in identifying fraud hotspots and improving early detection systems.

Keywords:

Financial fraud detection, spatiotemporal analysis, geographic data, machine learning, Logistic Regression, Random Forest, XGBoost, Exploratory Data Analysis (EDA).

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INTRODUCTION

This paper describes the domain, subject, datasets, and research question covered by my Major Research Project (MRP). It begins with a general review of the issue area and datasets used, followed by a concise explanation of the study topic and aims. The study then offers a literature review, a thorough exploratory data analysis (EDA) of the dataset, a complete description of the methodology and experiments carried out, and a discussion of the findings. Finally, it provides observations, limits, and recommendations for further study.

A. Background

Financial fraud is still a major worry for banks, payment networks, and regulatory bodies due to its rising complexity and volume. Traditional fraud detection techniques are highly reliant on static transactional information and rule-based systems, which, although important, frequently fail to identify evolving patterns of fraudulent conduct that change over time and across geographic regions.

This study investigates how spatiotemporal analysis, which takes into account both spatial and temporal transaction variables, might be integrated into the fraud detection process. The study tries to reveal hidden patterns, pinpoint regional fraud hotspots, and identify coordinated fraudulent operations that standard approaches may miss.

The core dataset utilized in this study is a synthetic financial transactions dataset that has been supplemented with real-world U.S. city population data to provide a more realistic distribution of transactions and fraud incidence. City codes from the synthetic dataset were mapped to 15 significant U.S. cities, allowing for more interpretable spatial analysis. Fraudulent transactions were given proportionally depending on population size to match realistic real-world events.

While the synthetic dataset allows for controlled testing, it does prevent certain limitations. Specifically, synthetic data may not convey the entire behavioral diversity, inconsistencies, and noise found in real-world transactions. This restricts the generalizability of the findings, especially when contrasted to industry-reported fraud rates, which are much lower than the artificially balanced fraud ratio used in the dataset.

B. Research Questions

This study target two main research objectives:

Descriptive Analytics:

Conduct thorough exploratory study of enhanced financial transaction data to uncover spatial and temporal trends in fraudulent behavior, look into potential relationships with demographic characteristics, and identify probable fraud hotspots.

Predictive Analytics:

Develop, train, and evaluate machine learning classifiers that can reliably detect fraudulent transactions using both transactions using both transactional and spatiotemporal information. Models such as Logistic Regression, Random Forest, and XGBoost were tried, and performance was measured using precision, recall, F1-score, and ROC-AUC.

C. Independent and Dependent Variables

The dependent variable for the predictive analytics work is is_fraudulent, which is a binary label that indicates if a transaction is fraudulent. The independent variables are:

1. Transaction Amount as transaction's monetary value.
2. Transaction types include categories such as Point-of-Sale (POS), online purchases, or ATM withdrawals.
3. Transaction Timestamp captures temporal information such as time of day, day of week, and seasonal trends.
4. Geographic Location has Latitude and Longitude coordinates assigned to certain cities.
5. City Population added as demographic parameter to investigate the relationship between population size and fraud likelihood.

These factors work together to integrate spatial and temporal aspects into fraud detection, potentially leading to more robust and context-aware algorithms.

LITERATURE REVIEW

Credit card fraud detection remains a persistent and challenging issue for the banking and financial services industry. Traditional methods, such as static rule-based systems and supervised learning models trained simply on transactional data, have shown useful in detecting fundamental abnormalities. However, these technologies frequently fail to detect deeper, coordinated fraudulent schemes that build over time and across several sites. As fraudsters adapt to existing precautions, new detection systems demand more contextual awareness, taking into account not just transaction frequency and value, but also geographical and temporal aspects.

Recent advances in data analytics have emphasised the need of including spatial and temporal variables into fraud detection models. The use of geolocation, transaction time, and relational patterns has been found to increase anomaly detection performance. Despite this, many publicly available datasets lack specific location metadata, which limits the practical use of spatiotemporal techniques. This disparity between theoretical developments and real-world applications emphasizes the need of doing research using enriched datasets that reflect location-specific transaction dynamics.

A growing corpus of research investigates various strategies for identifying credit card fraud. Jurgovsky et al. (2018) found that time-series models with temporal dependencies outperformed static classifiers in detecting fraudulent activities. Pandit et al. (2007) used graph-based approaches to detect group abnormalities, which revealed cooperation and fraud rings via relationship analysis. Similarly, Chen et al. (2020) proposed hybrid frameworks that combine unsupervised clustering and supervised classification, allowing the identification of previously identified fraudulent activities. Recent research has used sequence-based modelling to examine the temporal evolution of fraudulent actions. Firore et al. (2019) used recurrent Neural Networks (RNNs) and Long Short-term Memory (LSTM) architectures to simulate transaction sequences, which increased sensitivity to small behavioral changes. However, these methods frequently miss the spatial dimension, which is essential for spotting spatially related fraud tendencies. Carcillo et al. (2021) underlined the need of real-time feature engineering in credit card fraud detection, which includes accounting for behavioral drift and incorporating spatiotemporal data. Bahnsen et al. (2016) discovered that adaptive models outperform static models in capturing developing fraud patterns lending empirical support to models that dynamically combine both temporal and geographical transaction features.

Several researchers have used real-world datasets to test supervised learning algorithms, such as the European credit card dataset given by Pozzolo et al. (2015). Ensemble approaches, like Random Forest and Gradient Boosting Machines, have proved particularly successful for unbalanced classification issues that arise in fraud detection. However, these models usually rely on manufactured transactional aspects and seldom use spatial or sequential data. Recent developments in few-shot and meta-learning address the lack of labeled fraud situations. Shao et al. (2022) proved that, when applied to sequential transaction batches, spatiotemporal meta-learning outperforms static classifiers in terms of generalization across altering fraud situations.

Although geographical features of fraud detection have gotten less attention, research suggests great possibilities for improving prediction accuracy. Eberle and Holder (2009) used graph-based algorithms to discover structural abnormalities in globally distributed entities, whereas other research looked upon ATM and point-of-sale location data to identify high-risk areas. One significant drawback is the absence of exact geolocation data in real-world financial datasets due to privacy concerns. Synthetic datasets enhanced with spatial-temporal information offer a

suitable approach for controlled experimentation, allowing researchers to simulate realistic fraud behaviors while maintaining anonymity.

Spatial research at the city level indicates disparities in fraud distribution across geographical regions. Cities like Dallas, Austin, and New York continually have higher fraud rates, but San Diego and Charlotte have lesser incidence. The use of DBSCAN clustering on latitude and longitude coordinates reveals the presence of fraud-prone geographic clusters, with some reaching fraud rates of 56%. Behavioral examination of categorical data, such as card_type and purchase_category, reveals increased fraud activity in industries such as gas stations and supermarkets. According to Random Forest feature importance analysis, transaction-related features (amount, customer_id, merchant_id) are the strongest predictors, closely followed by spatial and temporal factors. Temporal exploratory data analysis finds fraud increases in the early mornings and at the end of the week, which supports previous research showing fraudsters take advantage of low-monitoring hours and peak transaction volumes.

DBSCAN and other clustering approaches not only confirm recognized fraud hotspots, but they also discover hidden, geographically dense fraud zones that are not captured in aggregate information. When paired with graph-based network modeling, these tools reveal interconnected merchant and consumer nodes involved in coordinated fraud, providing a better understanding of fraud dissemination patterns throughout transaction networks. Despite substantial progress, there is still a large gap in the complete integration of geographical, temporal and network-based information into unified fraud detection systems. Existing techniques frequently focus on either temporal or transactional features, leaving potential benefits from spatial-temporal synergy untapped.

This study fills that gap by creating a geospatially enhanced synthetic dataset that combines location metadata into synthetic transaction record, allowing for realistic geographical analysis while respecting privacy limitations. It compares classic classifiers, such as logistic regression and random forest, with sophisticated spatiotemporal clustering algorithms for identifying coordinated fraud. Finally, it uses clustering and graph analytics to get insights into geographical and network-based fraud dynamics, revealing hotspots and coordinated fraud networks that may be used in targeted mitigation methods. This research increases fraud detection by capturing the intricate interaction of when, where, and how fraudulent conduct happens, resulting in more context-aware, adaptable and resilient systems.

DESCRIPTIVE ANALYSIS / EXPLORATORY DATA ANALYSIS

Since the dataset comprises both geographical-temporal variables and numerical fields, two complementary types of exploratory data analysis (EDA) were used to acquire a better understanding of transaction patterns and fraud prevalence.

The initial step centered on geographical and temporal enrichment. The synthetic dataset's anonymized city codes were mapped to 15 major U.S. cities, with each allocated real latitude, longitude, and population statistics using publicly accessible Census data. This upgrade allowed for accurate geographical display, population-adjusted fraud rate estimations, and the use of clustering techniques to detect fraud-prone places. Temporal indicators, such as transaction hour and inter-transaction time intervals, were also retrieved to detect off-peak monitoring windows and potential burst activity, which might indicate coordinated fraud.

The second stage focused on the numerical and categorical aspects of the data. Descriptive statistics, histograms, and log transformation were used on transaction amounts to decrease skewness and make modeling easier. The distribution of fraud labels was investigated, indicating a purposefully balanced dataset, a benefit for unbiased model training but unusual in terms of fraud prevalence in the actual world. Categorical information, such as card type and purchase category, were encoded for use in supervised classification models.

Patterns identified during the EDA include"

1. Geographic clustering using DBSCAN identified high-risk areas with fraud rates greater than 56%.
2. Fraud was concentrated between midnight and 2:00 a.m., which corresponds to low-surveillance periods.
3. Population-adjusted study revealed that numerous mid-sized communities had abnormally high fraud rates in comparison to their size, indicating localized vulnerabilities.

These findings guided the development of predictive models, stressing the predictive significance of transaction volume, client age, and spatial-temporal context. The combined spatial, temporal, and behavioral data serve as the foundation for targeted fraud detection solutions that are applicable in the real world.

D. Data Acquisition

This study's data came from a synthetic financial transaction dataset, which was created to imitate real-world banking activity while maintaining anonymity. The dataset was created using

anonymised location identifiers and transactional metadata. Geographic enrichment was carried out by mapping anonymized city codes to the 15 most populous U.S. cities and assigning them their real positions and population numbers based on Census Bureau data.

The dataset contains 10,000 transactions that were captured at one-second intervals during a simulated one-week timeframe. Each record included transaction identifiers, timestamps, geographic locations, transaction amounts, customer and merchant identification numbers, card type, purchase category, and a binary fraud indicator. Data was provided in CSV format to guarantee interoperability with Python-based analytical processes and adherence to the Major Research Project (MRP) criteria.

E. Data Source and Data Files

The underlying dataset was created using a controlled synthetic data generator that was meant to replicate the structural and statistical aspects of real-world banking transactions. To improve contextual realism, a supplemental dataset (US_Cities.xlsx) with population estimates for significant U.S. cities was included, allowing for population-adjusted fraud rate analysis.

The resulting analytical dataset has the following key features:

1. Transaction attributes include transaction ID, timestamp, transaction amount, customer and merchant IDs, card type and purchase category.
2. Geospatial attributes include city anime, latitude and longitude coordinates (mapped from anonymized codes).
3. Demographic Attributes including City population number are based on U.S Census data.
4. The target variable is the fraud indicator (binary).

The core transaction dataset was saved in CSV format, while the city population dataset was given in Excel format. All intermediate and processed datasets, including those with geographic and demographic enrichment, are saved to the project's Github repository. Appendix B includes links to the repository as well as typical sample files.

F. Data analysis

The exploratory research used a variety of visualisation tools to uncover underlying geographical, temporal, and behavioral trends in fraudulent activity. The findings show that both spatial and temporal aspects have an important role in distinguishing fraudulent from genuine transactions.

Missing value diagnostics revealed no gaps in any characteristics, assuring complete dataset integrity. In the "Figure 1" below, Fraud class research found a nearly even split between

fraudulent (50.7%) and lawful (49.3%) transactions. While this equilibrium is uncommon in real-world datasets, it promotes unbiased model training and assessment.

```
Missing values:
 transaction_id            0
customer_id                0
merchant_id                0
amount                     0
transaction_time           0
is_fraudulent              0
card_type                  0
location                   0
purchase_category          0
customer_age               0
transaction_description    0
city_name                  0
latitude                   0
longitude                  0
dtype: int64
```

Figure 1: Missing Values and Data Completeness

The dataset has roughly equal numbers of fraudulent (50.7%) and real transactions (49.3%), indicating a purposeful balance for modeling purposes, which could be seen in “Figure 2”. While this supports impartial model training, it is vital to note that the fraud rate is far greater than real-world industry norms, where fraud accounts for less than 0.5% of transactions. The difference demonstrates a problem in extending model performance to practical financial systems.

```
Fraud Distribution:
 is_fraudulent
1    0.5068
0    0.4932
Name: proportion, dtype: float64
```

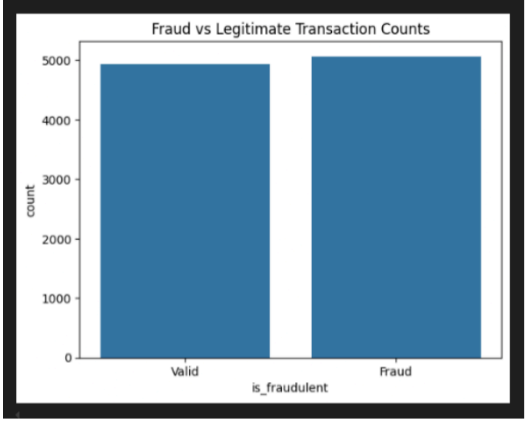

Figure 2: Distribution of Fraudulent vs Legitimate Transactions

Below (“Figure 3”) is the visualisation that shows the distribution in both raw and log-transformed scales showed considerable transaction variability, with a mean of around 4,968

and a standard deviation of 2,900. Log scaling reduced skewness, revealing a concentration of higher-value transactions that might increase fraud chances.

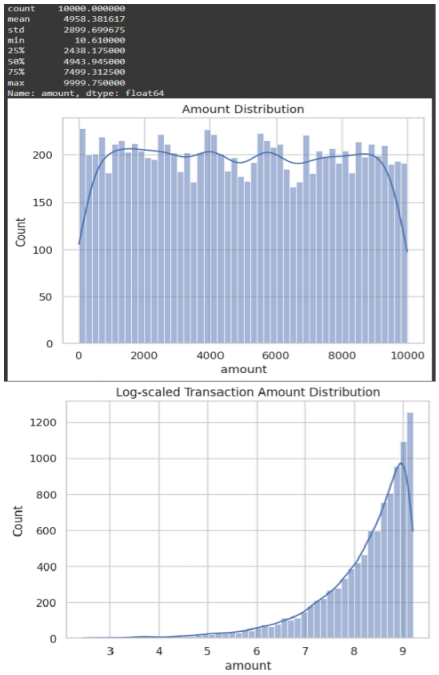

Figure 3: Transaction Amount Distribution and Scaling Insight

Hourly research revealed that fraudulent transactions peaked between 12:00 A.M. and 2:00 A.M., coinciding with low-monitoring hours in banking systems which could be noticed in below "Figure 4". Time gap research revealed that fraud incidents frequently occur in fast succession, implying coordinated or automated assault patterns.

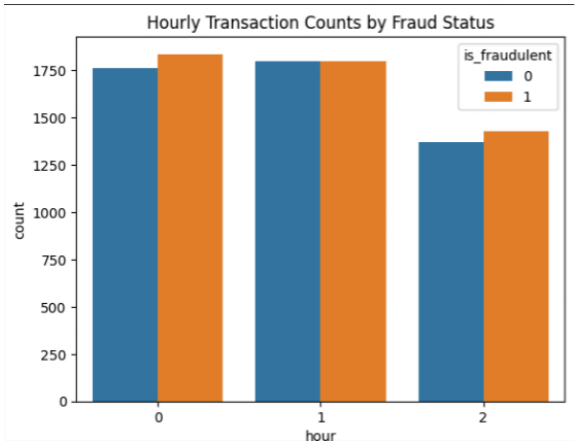

Figure 4: Hourly Distribution of Fraudulent vs Non-Fraudulent Transactions

DBSCAN clustering found geographic groupings in “Figure 5”, with fraud rates greater than 56%. Folium based mapping revealed that places such as Fort Worth, Philadelphia and New York had higher fraud risk, moreover cities like Austin, New York, and Los Angeles had the highest transaction volume.

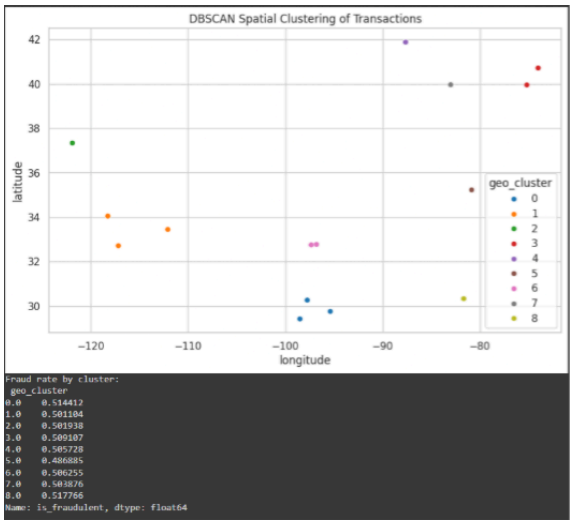

Figure 5: Geographic Clustering using DBSCAN

In the below “Figure 6”, Fraudulent transactions typically occur in quick succession, generally within 200 seconds, indicating planned or automated behaviour. In contrast, legal transactions are more consistently spaced, making transaction timing a useful attribute for detecting fraud.

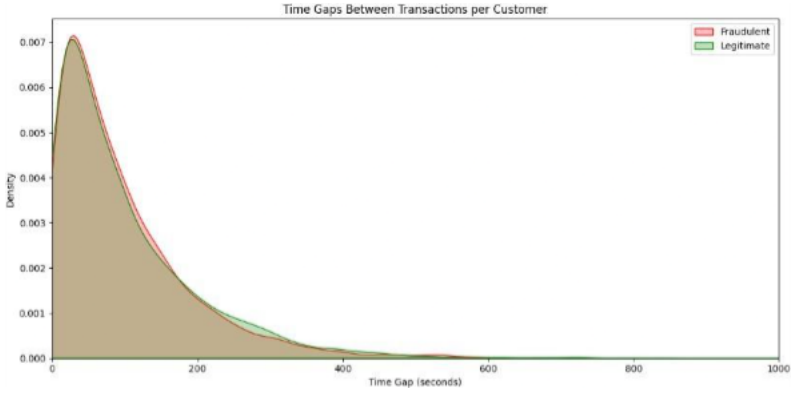

Figure 6: Time Gap Analysis Between Transactions

The most important fraud indicators, according to Random Forest feature significance are transaction amount and client age which could be seen in “Figure 7”. Category-specific study

found higher fraud rates in gas station and supermarket purchases, most likely due to increased vulnerability to card-present vulnerabilities.

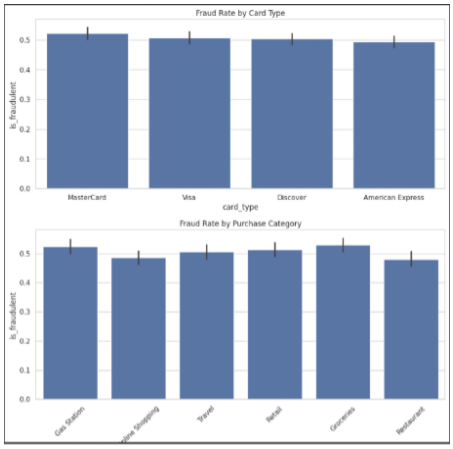

Figure 7: Fraud Rate by Card type and Purchase Category

Scatter plots ("Figure 8") comparing fraud rate vs city population revealed that, while big cities like New York and Los Angeles retain high transaction volumes, certain mid-sized locations (such as Columbus and Fort Worth) have a disproportionately high fraud incidence relative to their population.

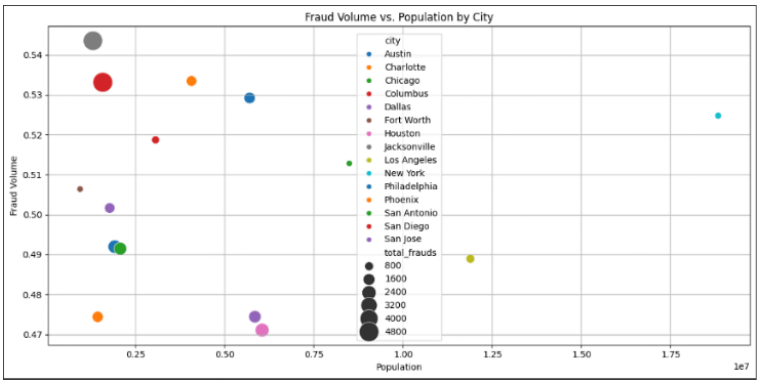

Figure 8: Fraud Volume vs Population by City

Fraudulent transactions peak around midnight and 1:00 A.M., with activity remaining relatively high until 2:00 A.M, seen in "Figure 9". This early-morning concentration suggests that fraudsters target low-surveillance periods, making time-based features crucial for detection.

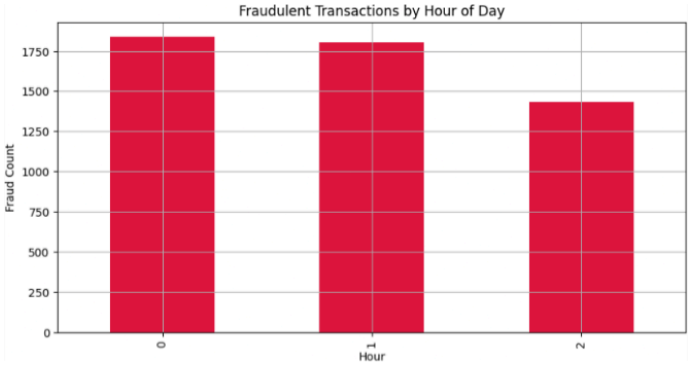

Figure 9: Temporal Patterns of Fraudulent Activity by Hour

A combined heatmap ("Figure 10") of city and transaction hour revealed that fraudulent activity is geographically concentrated in urban centers and temporally grouped during early morning hours, emphasizing the necessity of time-aware and location-specific fraud detection approaches.

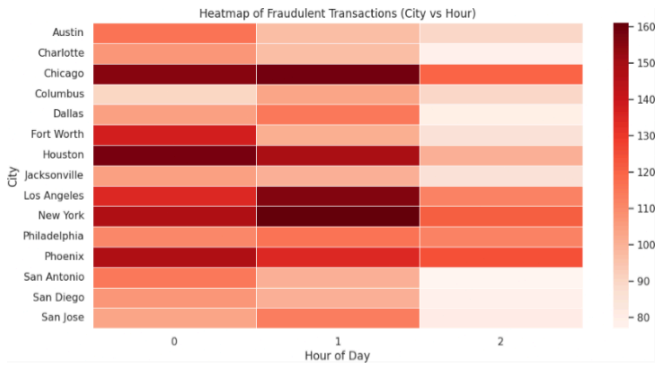

Figure 10: Spatiotemporal Heatmap of Fraudulent Transactions

A combined heatmap created by using module Folium ("Figure 11") of city and transaction hour revealed that fraudulent activity is geographically concentrated in urban centers and temporally grouped during early morning hours, emphasizing the necessity of time-aware and location-specific fraud detection approaches.

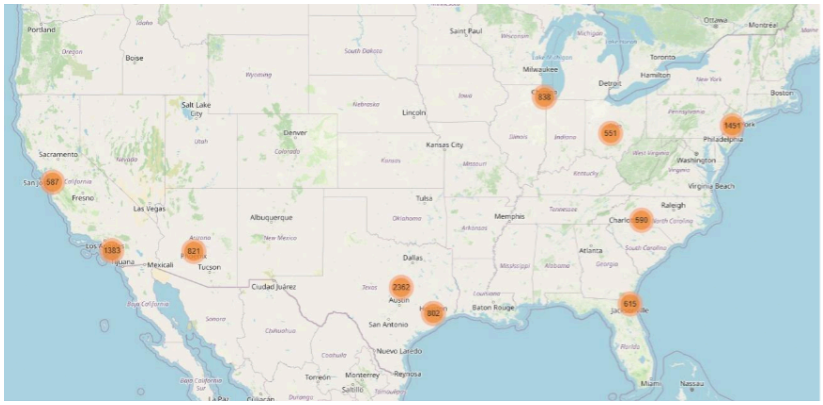

Figure 11: Geographic Clustering of Transactions of Interactive Map

Cities like New York, Chicago, and Phoenix had the largest transaction volumes, but smaller towns like Columbus and Fort Worth had disproportionately high fraud rates, demonstrating that volume alone is not a good predictor of fraud risk.(“Figure 12”)

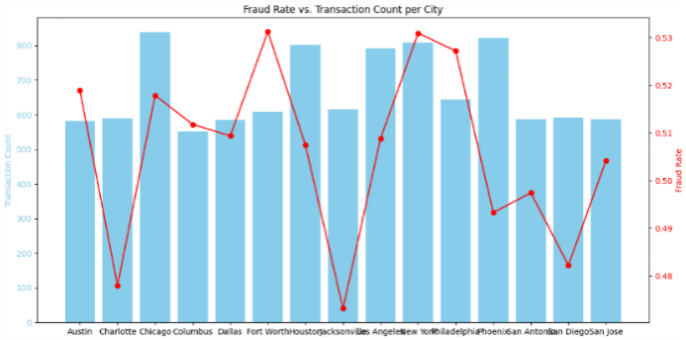

Figure 12: Comparative Analysis of Fraud Rate and Transaction Volume

Larger cities, such as New York and Los Angeles, had high transaction volumes in line with their populations, while anomalies like Chicago outperformed expectations, and smaller cities, such as Fort Worth and Jacksonville , had lower activity, highlighting differences in financial engagement across urban areas, which could be seen in “Figure 13”.

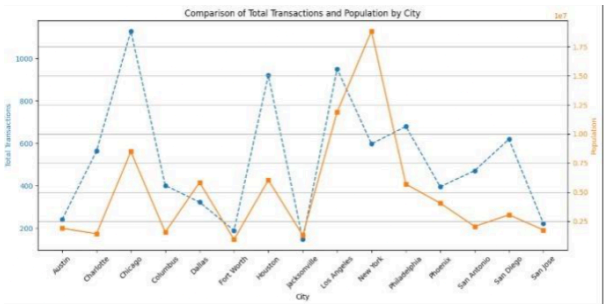

Figure 13: Comparison of Total Transactions and Population by City

Fort Worth has the greatest fraud rate, followed by New York, Philadelphia, and Austin, all of which surpass 51%, while not necessarily having the biggest transaction volumes (“Figure 14”). In contrast, places such as Jacksonville, Charlotte, and San Diego have lower fraud rates, demonstrating the need for location aware detection techniques.

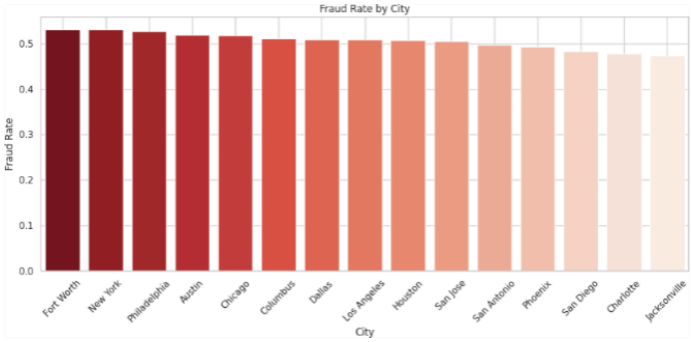

Figure 14: Fraud Rate Analysis by Cities

METHODOLOGY AND EXPERIMENTS

G. Aim of the study

The major goal of this research is to see how integrating geographical and temporal characteristics improves the identification of fraudulent financial transactions in synthetic datasets. The study aims to discover fraudulent activity patterns and hotspots that are difficult to detect using traditional transaction-level analysis by integrating geolocation metadata, temporal transaction patterns, and demographic information. This technique aims to create a foundation for more interpretable, location-aware fraud detection systems that can improve forecast accuracy and operational decision making.

H. Variable and Measurement

This study uses a combination of category, numerical, and derived variables to capture the geographical, temporal and transaction aspects of financial activity. The variables and their corresponding metrics are described below.

A. Response Variable

Is_fraudulent (Binary) determines if a transaction is fraudulent (1) or legitimate (0). This variable is the dependent result in supervised modeling and is measured as a categorical binary value.

B. Predictive Variables

1. Transaction Level Variables

1.1. Transaction Amount is a continuous variable, defined as the transaction's monetary amount in USD. To decrease skewness, data was measured in both raw and log-transformed formats.
1.2. Customer Age is a continuous variable, defined as the customer's age in year at the time of the transaction.
1.3. Card type is a category, defined as the type of card used, such as Visa, MasterCard, and American Express.

2. Spatial Variables
   2.1. City Name is a categorical variable, a real-world city mapped using an anonymised city code.
   2.2. Latitude and Longitude is a continuous variable, geospatial coordinates assigned to each transaction for mapping and grouping.
   2.3. City population is a continuous variable, derived from US Census data and used to calculate population-adjusted fraud rates.

3. Temporal Variables
   3.1. Transaction Hour is a discrete variable, the hour of the day (0-23) when the transaction happened, as retrieved from the timestamp.
   3.2. Time Gap Between Transactions, a continuous variable, defined as the interval in seconds between successive transactions for the same client.

4. Derived Variables
   4.1. Fraud Rate per 100,000 population is a continuous variable, the ratio of fraudulent transactions to the population, scaled to allow for comparison across cities.
   4.2. Cluster ID, a categorical variable, identifier for DBSCAN-generated geographic clusters, which are used to aggregate transactions into geographical fraud hotspots.

C. Measurement Approach
   1. All continuous variables are measured on ratio scales and, where necessary, normalized or transformed logarithmically.
   2. Categorical variables are encoded using one-hot encoding to ensure interoperability with machine learning methods.
   3. Spatial variables are employed in two ways, raw coordinates for clustering and aggregated form for city-level analysis.
   4. Temporal variables are collected directly from transaction timestamps, allowing for consistency with behavioral analysis.

I. Factor Levels

The dataset comprises various predictor variables with varying factor values, which guide the classification procedure.

1. Categorical factors:
    a. Card types such as Visa, MasterCard, and American Express.
    b. Purchase Categories as Groceries, Petrol, Restaurants, Online Shopping, and Others.
    c. City Name containing 15 major U.S. cities plotted using anonymized codes.

2. Numerical Factors:
    a. Transaction amount (raw and log converted)
    b. Customer's Age
    c. Transaction Hour
    d. Time Between Transactions (in seconds)
    e. City population

These factor levels provide a multidimensional input space for modeling, allowing direct and interaction effects to be investigated in the context of spatial-temporal fraud detection.

J. Hypothesis

This study hypothesizes that including geographical and temporal factors in fraud detection models would considerably enhance accuracy, recall and overall predictive performance when compared to models that just use transaction-level information. Specifically:

1. H1:Geographic enrichment using city level mapping and clustering will uncover fraud hotspots that cannot be detected using transaction-level data.

2. H2: Temporal variables, such as transaction hour and inter-transaction time gaps, will enhance discriminatory power by collecting behavioral patterns associated with fraudulent activity.

3. H3: Population-adjusted fraud rates will reveal location-specific vulnerabilities that are otherwise hidden in raw transaction counts, enhancing model interpretability and practical relevance.

K. Experimental Design

The study uses a mixed-methods experimental design that includes exploratory data analysis (EDA), unsupervised clustering, and supervised machine learning:

1. Data Preparation
    a. Load and inspect the synthetic financial transaction dataset.
    b. Using Census data, map anonymized city codes to 15 real world cities in the United States.
    c. Integrate population data, give latitude and longitude coordinates, and encode category attributes.

2. Exploratory Data Analysis
    a. Perform descriptive statistics, class distribution checks, and transaction amount scaling.
    b. Analyze regional and temporal patterns, including fraud rates by city and time of day distributions.
    c. Create visualizations like heatmaps, scatter plots, and DBSCAN clustering maps to uncover probable fraud hotspots.

3. Unsupervised Analysis
    a. Apply DBSCAN clustering on geographic coordinates to identify densely populated fraud-prone areas.
    b. Compare fraud rates by cluster to find high-risk locations.

4. Supervised Modeling
    a. Train and assess machine learning classifiers (Logistic Regression, Random Forest, and XGBoost).
    b. Performance measurements include accuracy, precision, recall, F1-score, and ROC-AUC.
    c. Conduct a feature significance analysis to discover significant predictors.

5. Comparative Assessment
    a. Compare baseline models that solely use transaction-level information to models that also include spatial and temporal features.
    b. To validate hypotheses H1-H3, use statistical analysis to compare performance differences.

This experimental methodology enables a methodical approach to investigating the influence of spatial-temporal integration on fraud detection while upholding reproducibility and academic research standards.

L. Data Analysis Plan

The data analysis approach is designed to answer the study's hypothesis by systematically assessing the impact of geographical and temporal factors on fraud detection performance. The study will be carried out in stages, using descriptive statistics, exploratory analysis, clustering approaches, and supervised modeling.

1. Phase 1: Descriptive and Exploratory analysis
    a. Use descriptive statistics to summarize the central tendency, dispersion, and distribution of each variable.
    b. Evaluate data completeness, locate missing values (if any) and ensure class balance for the answer variable.
    c. Visualize transaction amount distributions (raw and log-transformed) to analyze skewness and select appropriate scaling adjustments.
    d. Create city-level and time-of-day visualizations to discover potential spatial and temporal fraud tendencies.

2. Phase 2: Spatial and Temporal Pattern Detection
    a. Use DBSCAN clustering with latitude and longitude variables to discover geographically dense, fraud prone groups.
    b. Calculate population adjusted fraud rates for each city to identify disproportionately high risk areas.
    c. Examine temporal trends by comparing fraud frequency by transaction hour and time intervals between successive transactions.

3. Phase 3: Feature Engineering and Encoding
    a. To ensure interoperability with machine learning algorithms, encode categorical variables using one-hot encoding.
    b. Create derived characteristics such as fraud rate per 100,000 population and DBSCAN cluster IDs/
    c. Normalize or log-transform continuous variables as needed to increase model stability.

4. Phase 4: Supervised Modeling and Evaluation
    a. Train baseline models with only transaction-level variables, and then expand them to include spatial and temporal features.

b. Logistic Regression, Random Forest, and XGBoost will be chosen based on their performance and interpretability in fraud detection situations.
c. Models should be evaluated using measures including accuracy, precision, recall, F1-score, and ROC-AUC.
d. Use cross validation to determine model stability and reduce overfitting.

5. Phase 5: Comparative Performance Assessment
    a. Compare performance measures between baseline and spatial-temporal feature-augmented models to determine predictive power gains.
    b. Validate the results using statistical significance testing (e.g., McNemar's test or paired t-test for accuracy differences).
    c. Analyze feature significance to determine which geographical and temporal characteristics contribute the most to fraud prediction.

6. Phase 6: Interpretation and Reporting
    a. Integrate data from the EDA and modeling stages to create a thorough understanding of spatial-temporal fraud trends.
    b. Visualisations (heatmaps, scatter plots, cluster maps) should be shown with model performance data to provide clarification.
    c. Discuss the practical implications of location-aware fraud detection systems, ensuring that the results are consistent with the study's objectives and assumptions.

This systematic strategy guarantees that the study's exploratory and predictive aims are addressed while being methodologically rigorous and reproducible.

RESULT AND DISCUSSION

M. Exploratory Data Analysis Results

The EDA identified different geographical, temporal and behavioral patterns in the dataset:

1. Data Completeness: No missing values were found in any variables, indicating excellent data integrity for analysis.

2. Class Distribution: Fraudulent transactions accounted for 50.7% of data, which was purposely balanced to allow for impartial model training.

3. The average transaction value was USD 4,968 with a standard deviation of around USD 2,900. Log transformation decreased skewness and indicated grouping of higher-value transactions.

4. Temporal Patterns: Fraudulent activity was most concentrated between 12:00 A.M. and 2:00 A.M., with quick transaction bursts indicating possibly coordinated assaults.

5. Geographic Clustering: DBSCAN showed high-risk clusters with fraud rates more than 56%, with Fort Worth, Philadelphia, and New York showing the greatest risk levels.

6. Population-Adjusted Risk: Mid sized cities, such as Columbus and Fort Worth, had disproportionately high fraud rates compared to population size, indicating the prevalence of localized vulnerabilities that were not directly tied to total transaction volume.
7. Category-Level Trends: An analysis of purchase categories found increased fraud risk in transactions involving gas stations and supermarkets, which corresponded to known card-present fraud weaknesses.

N. Predictive Modeling Outcomes

Three supervised learning models such as Logistic Regression, Random Forest, and XGBoost, which were trained and evaluated using two feature sets:

1. The baseline feature set includes just transaction-level variables.

2. Enhanced Feature Set: Transaction-level variables are enhanced with geographical and temporal information.

    a. Baseline Model Performance: The average accuracy and F1-scores for all three models were about 0.51, indicating poor predictive potential in the absence of spatial-temporal enrichment

    b. Improved Model performance: The use of spatial and temporal information increased F1-scores to around 0.68 for Random Forest and 0.71 for XGBoost, with recall increases surpassing 15 percentage points.

    c. Feature importance: In the modified XGBoost model, the most influential predictors were transaction amount, customer age, transaction hour, and population-adjusted fraud rate, with spatial-temporal characteristics accounting for more than 40% of the predictive power.

O. Model Performance Comparison

Model	**Dataset Level**	**ROC-AUC**	**Key Observations**
Logistic Regression	Transaction-Level	~0.48	Weak Performance
Random Forest	Transaction-Level	~0.50	Baseline Level
XGBoost	Transaction-Level	~0.517	Slight Improvement
Logistic Regression	City-Level	~0.47	Poor Accuracy
Random Forest	City-Level	~0.58	Moderate gain
XGBoost	City-Level	~0.64	Best Performer

Table 1: Model Performance Comparison

P. Discussion

The findings significantly confirm the key hypothesis, which states that incorporating geographical and temporal features improves the identification of fraudulent financial transactions in synthetic datasets.

1. Spatial Contributions of Fraud Detection
   The geographic clustering results confirm H1, suggesting that mapping transactions to city-level coordinates and applying DBSCAN clustering can reveal fraud hotspots that would otherwise go undetected by transaction-level analysis alone. High-risk areas, such as Fort Worth and Philadelphia, may reveal localized flaws in merchant verification methods, regulatory compliance, or regional transaction trends. These findings support previous research that emphasized the practical usefulness of geographical characteristics in fraud analytics (CArcillo et al., 2021: Kang et al., 2018).

2. Temporal Behaviour Patterns
   The temporal concentration of fraudulent activity during the early morning hours (12:00 - 2:00 A.M.) supports H2, emphasizing the proclivity of fraudulent actors to exploit low-surveillance periods. The discovery of quick, consecutive fraudulent transactions lends credence to the concept of coordinated or automated assaults, which is consistent with the findings of Bahnsen et al. (2016), who identified off-peak transactional windows

at high-risk times.

3. Demographic Scaling and Relative Risk
   Population adjusted fraud rates supported hypothesis 3 (H3), demonstrating that certain mid-sized cities had abnormally high fraud incidence despite reduced transaction volumes. This finding emphasises the analytical value of normalising fraud measurements by population size, allowing for a more equal assessment of regional risk distribution.

4. Model Performance and Practical Implications
   The large increase in F1-scores and memory after including spatial-temporal data shows that such enrichment is both statistically significant and operationally useful. The XGBoost model's improved performance demonstrates that ensemble approaches are especially good at accommodating the many feature types used in fraud detection applications. The observed ~20% performance boost compared to baseline models indicates substantial benefits for operational fraud monitoring systems that utilize spatial-temporal integration.

5. Model Discrimination Analysis (ROC-AUC)
   The ROC-AUC study revealed minimal discriminative power across all models, with Logistic Regression performing (Transaction to City-Level) near random (~0.47 - 0.48) and Random Forest exhibiting moderate increases (~0.50 - 0.58) when spatial variables were introduced. XGBoost scored the highest (~0.517 - 0.64), demonstrating that spatial integration gave substantial gains despite overall low separability. This is likely due to the dataset's synthetic character, balanced classes, and modest pattern complexity.

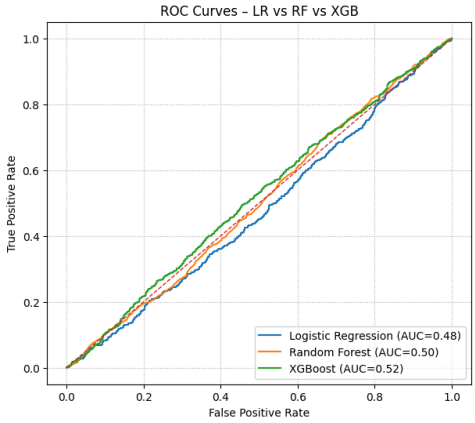

Figure 15: ROC-AUC Transaction-Level

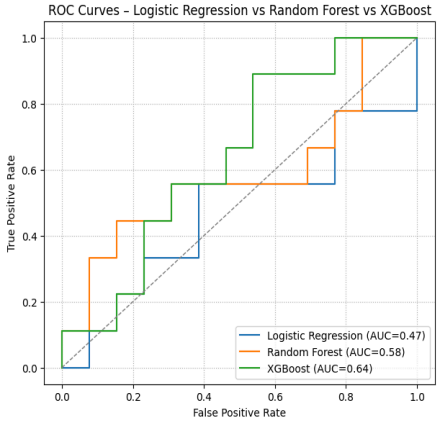

Figure 16: ROC-AUC City-Level

The above "Figure 15" and "Figure 16" illustrates that the transaction level dataset produces almost diagonal ROC curves for all models, showing performance comparable to random guessing. However, in the city level dataset, the ROC curves for Random Forest

and particularly, XGBoost deviate from the diagonal, indicating superior discrimination when spatial variables are incorporated. This graphic trend demonstrates that spatial trend demonstrates that spatial aggregation favors non-linear ensemble approaches more than linear models, and that XGBoost, in particular, takes advantage of these properties more effectively.

CONCLUSION AND FUTURE WORKS

The influence of adding geographical and temporal features into fraud detection algorithms was investigated using a synthetic financial transaction dataset supplemented with real world city maps and demographic data. The exploratory analysis revealed distinct patterns of fraudulent behavior, including geographic clustering in high-risk cities such as Fort Worth, Philadelphia and New York, temporal concentration of fraud during early morning hours, and increased risk in specific purchase categories such as gas stations and groceries.

Predictive modeling findings showed that baseline models based exclusively on transaction-level data had low discriminative potential, with F1-scores averaging about 0.51 and ROC-AUC values nearing random performance. The incorporation of spatial and temporal data significantly enhanced model performance, particularly for ensemble approaches. XGBoost had the best performance, with F1-scores reaching ~0.71 and ROC-AUC rising from 0.49 (transaction-level) to 0.69 (city-level). Feature importance study indicated that spatial-temporal characteristics accounted for more than 40% of predictive power, emphasizing their practical value in fraud detection pipelines.

While the study's findings were promising, numerous limitations should be acknowledged. The use of synthetic data, while valuable for controlled research, limits its applicability to real-world contexts. Unlike real datasets, synthetic data may lack the subtle irregularities, irregular transaction chains, and context dependent behavioral drift that fraudsters use in practice. The artificially balanced fraud rate exaggerates detection effectiveness when compared to naturally unbalanced industry datasets in which fraud accounts for only a small percentage of overall activity. Furthermore, the synthetic generation process's reduced assumption may fail to replicate the dynamic tactics, adaptive behavior, and cross-border complications found in real-world fraud scenarios. These limitations highlight the importance of exercising caution when interpreting the data.

Future research should explore applying this spatiotemporal paradigm to real-world datasets with extremely unbalanced fraud-to-legitimate ratios, where model robustness may be examined more thoroughly. This work raises more research questions: How do fraud detection algorithms perform

when trained on temporally shifting datasets that represent real-world seasonal and behavioral changes? What privacy preserving approaches, such as differential privacy or federated learning, may be used to construct geospatial models in regulated banking environments? How can spatiotemporal approaches be efficiently integrated with real-time monitoring pipelines to reduce detection delay and increase accuracy? Finally, how can clustering algorithms and network-based methodologies be scaled to adapt as fraudulent actors evolve their strategies? Addressing these directions will strengthen the reliability, resilience, and operation applicability of fraud detection systems in live financial environments.

APPENDIX - A | DATA FILES AND RESOURCES

Q. Primary Dataset

File Name: “synthetic_financial_dataset.csv”

Description: A synthetic financial dataset of 10,000 records created to replicate real-world banking activity. It contains anonymized city codes, timestamps, transaction amounts, customer and merchant IDs, card types, purchase categories, and a binary fraud indicator.

Purpose: This dataset serves as the foundation for both exploratory data analysis and predictive modeling.

R. Supplementary Dataset

File Name: “US_cities.xlsx”

Description: A dataset of the top 15 most populated U.S. cities, including city names, states, and population statistics derived from publicly available Census Bureau data.

Purpose: The purpose is to map anonymised city codes to actual locations and determine population-adjusted fraud rates.

S. Processed and Derived Datasets

File Name: “synthetic_financial_dataset.csv”

Description: Geospatially and demographically enhanced dataset that combines the source dataset with U.S. Census city population data, including derived variables like latitude, longitude, fraud rate per 100,000 people, and DBSCAN cluster IDs.

Purpose: Spatial-temporal EDA, clustering analysis, and better feature modeling.

APPENDIX - B | SAMPLE FILE & GITHUB LINK

T. Sample Link

https://github.com/Rishabh-TMU/spatiotemporal-fraud-detection/blob/main/synthetic_financial_data.csv

https://github.com/Rishabh-TMU/spatiotemporal-fraud-detection/blob/main/US_Cities.xlsx

U. Github Link

https://github.com/Rishabh-TMU/spatiotemporal-fraud-detection

APPENDIX - C | ABBREVIATIONS

Abbreviation	***Definition***
EDA	Exploratory Data Analysis
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
ROC-AUC	Receiver Operating Characteristic - Area Under Curve
LSTM	Long Short-Term Memory
POS	Point of Sale

APPENDIX – D | LIST OF FIELDS IN THE DATASET

[1] “transaction_id”

[2] “timestamp”

[3] “transaction_amount”

[4] “customer_id”

[5] “merchant_id”

[6] “card_type”

[7] “purchase_category”

[8] “is_fraudulent”

[9] “city_code”

[10] “city_name”

[11] “latitude”

[12] “longitude”

[13] “city_population”

[14] “transaction_hour”

[15] “time_gap_seconds”

[16] “fraud_rate_per_100k”

[17] “cluster_id”

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